



图像与视频处理

Image and Video Processing

第6讲：图像增强-空域滤波

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Institute of Artificial Intelligence and Robotics

兰旭光



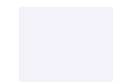
目录

- 强度转换
 - 图像反转
 - Log转换
 - Gamma 转换
 - 碎片化线性变换
- 直方图处理
 - 直方图均衡化
 - 直方图匹配
 - 局部直方图处理
- 空间滤波器
 - 空间相关性和卷积
 - 平滑的线性滤波器和空间掩码
 - 非线性滤波器
 - 锐化滤波器



锐化空域滤波器

- 基础
- 拉普拉斯算子
- 反锐化掩模和Highboost滤波
- 使用一阶导数进行非线性图像锐化--梯度





锐化空域滤波器: 基础

- 在强度（灰度值）上突出过渡
- 电子打印、医学影像、工业检测、自主引导;
- averaging-integration 平均-积分
- sharpen – differentiation 锐化-微分;

- 一维函数 $f(x)$ 的一阶导数为差分

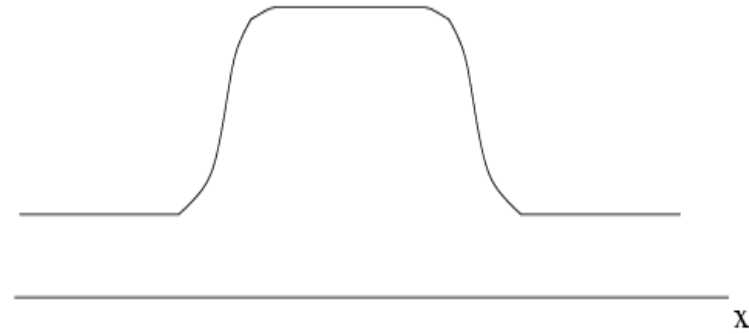
$$\frac{\partial f}{\partial x} = f(x+1) - f(x)$$

- 将 $f(x)$ 的二阶导数作为差分

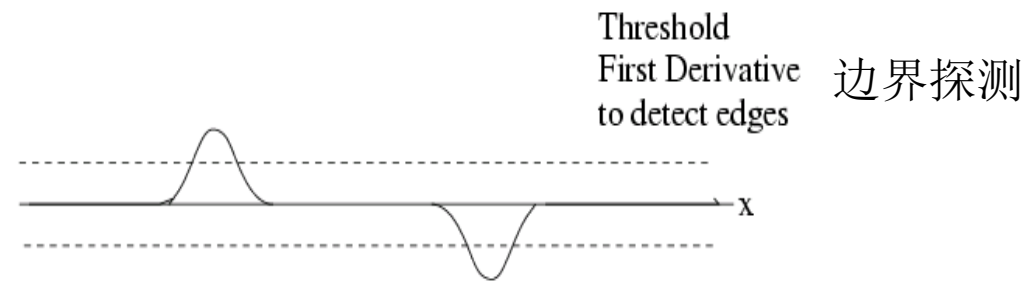
$$\frac{\partial^2 f}{\partial x^2} = f(x+1) + f(x-1) - 2f(x)$$



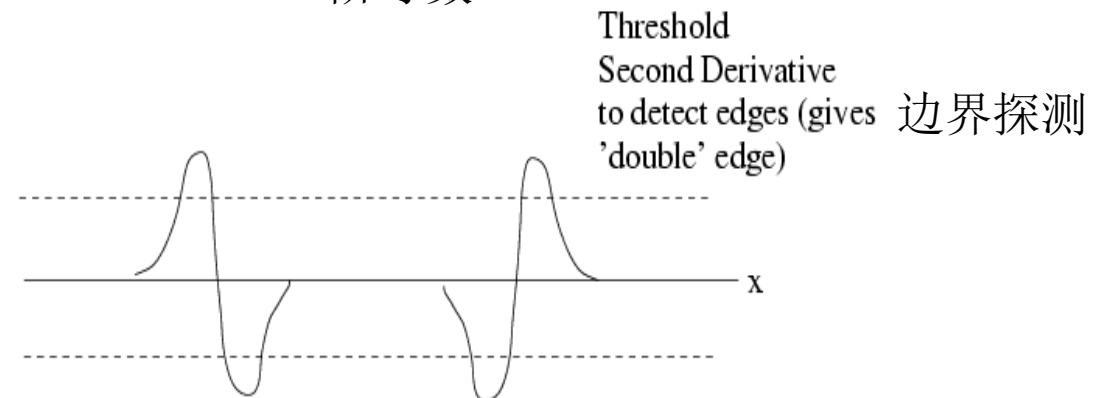
Image Trace



First Derivative 一阶导数



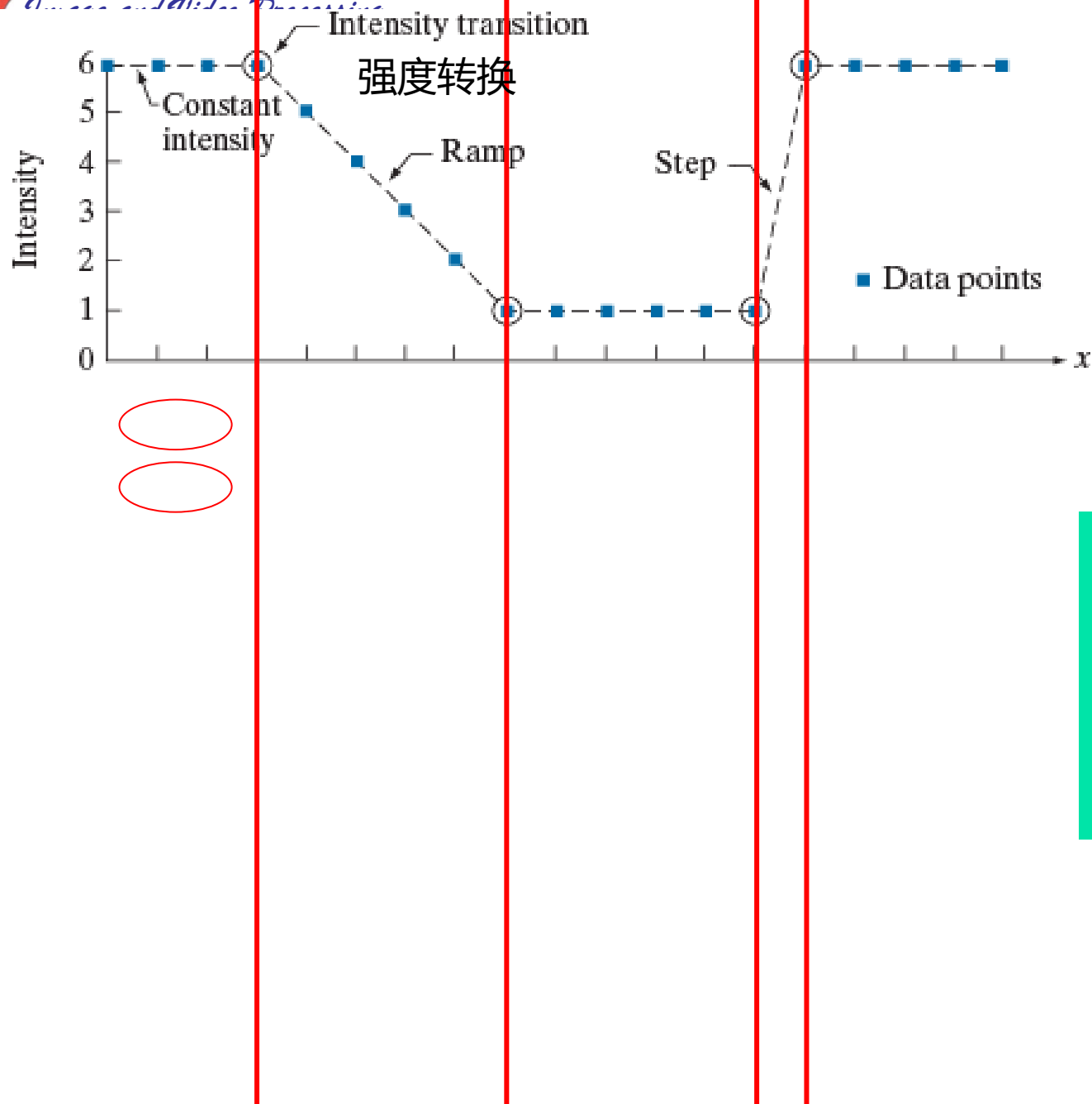
Second Derivative 二阶导数





图像与视频处理

Intensity and Gradient



$$\frac{\partial f}{\partial x} = f(x+1) - f(x)$$

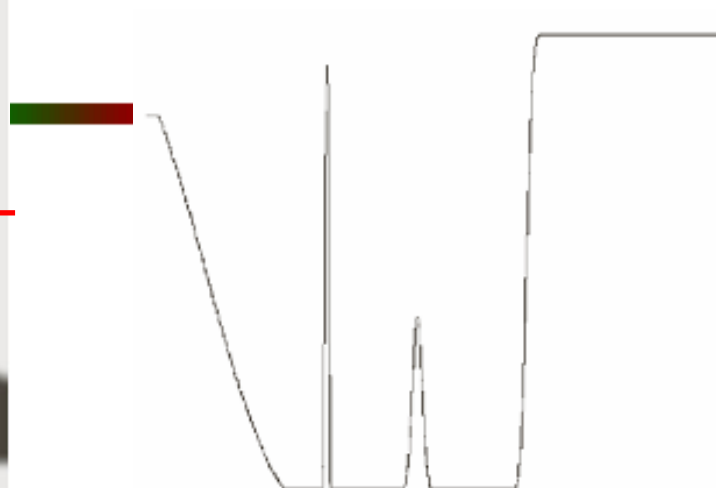
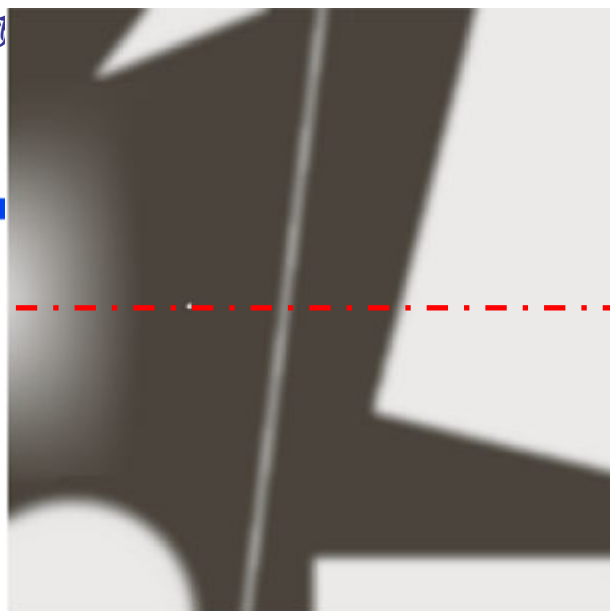
$$\frac{\partial^2 f}{\partial x^2} = f(x+1) + f(x-1) - 2f(x)$$

一阶导数:

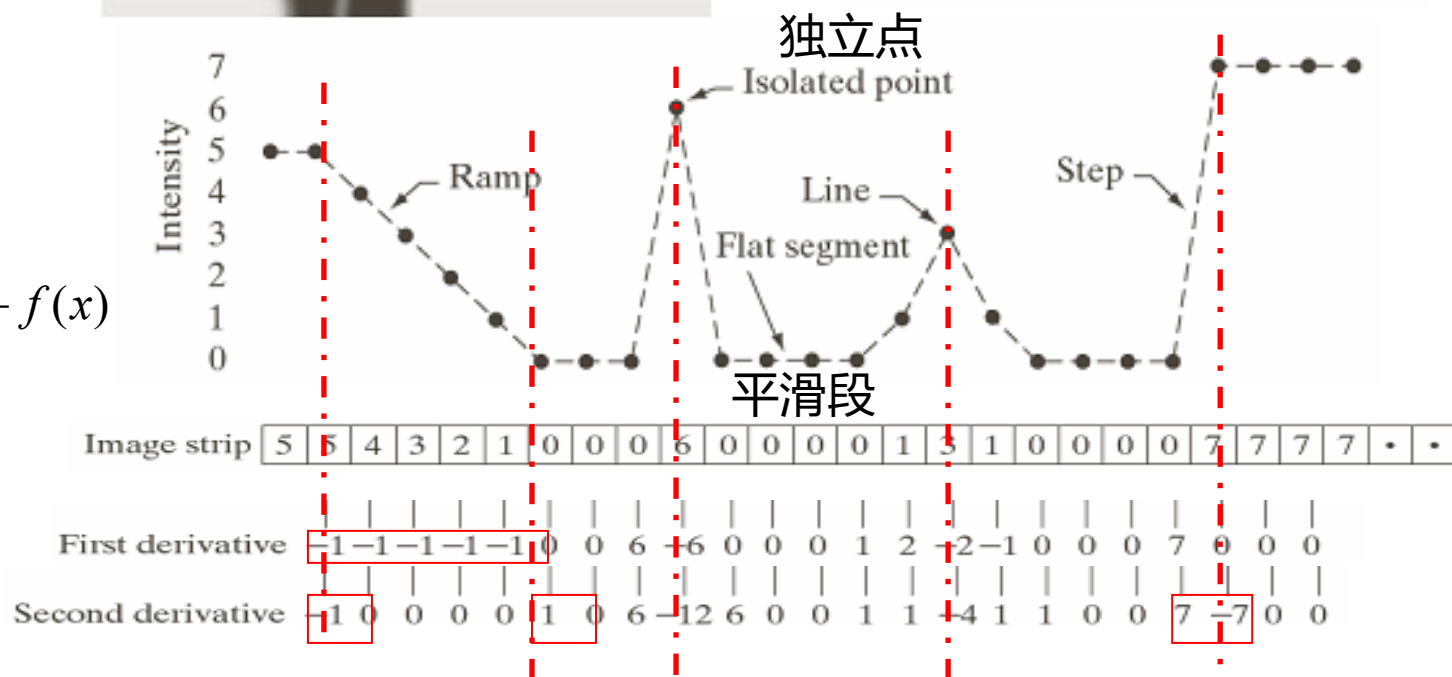
- 零: 常数
- 非零: 斜坡(ramp)或台阶(step)的起;
- 非零: 坡道

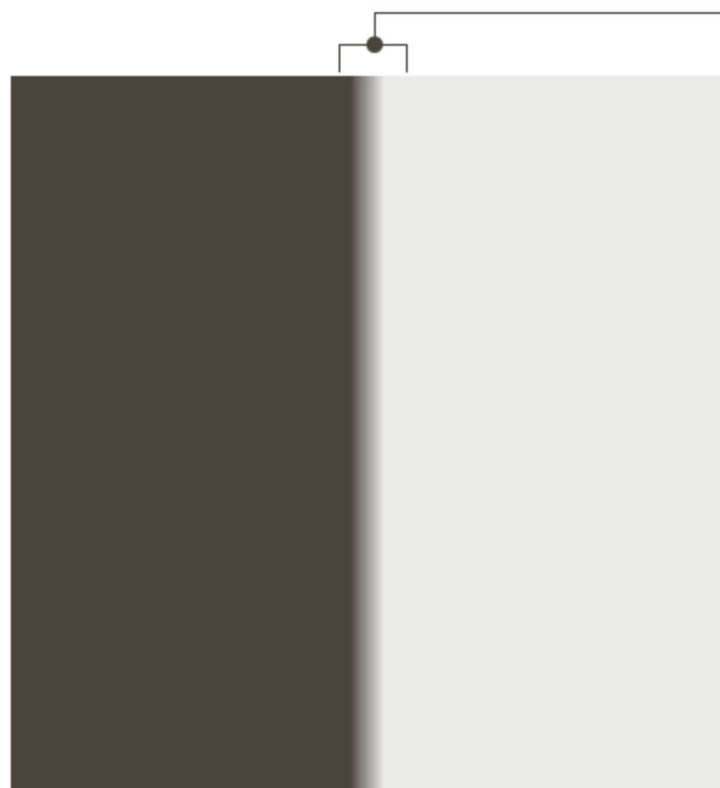
second derivative:

- 零: 常数
- nonzero: 斜坡(ramp)或台阶(step)的起始或结束;
- zero: 斜坡斜率固定



$$\frac{\partial f}{\partial x} = f(x+1) - f(x)$$





水平强度剖面

Horizontal intensity profile

First derivative

Second derivative

Zero crossing

a b

FIGURE 10.10

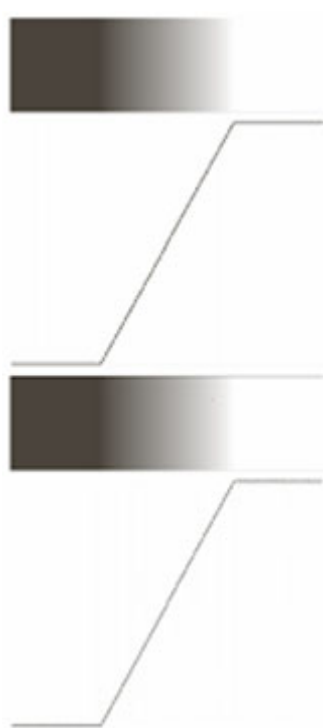
(a) Two regions of constant intensity separated by an ideal vertical ramp edge.
(b) Detail near the edge, showing a horizontal intensity profile, together with its first and second derivatives.

首先：检测一个点是否存在边缘
然后：标志：判断边缘像素是亮还是暗

Zero crossing(过零点): 边缘中心



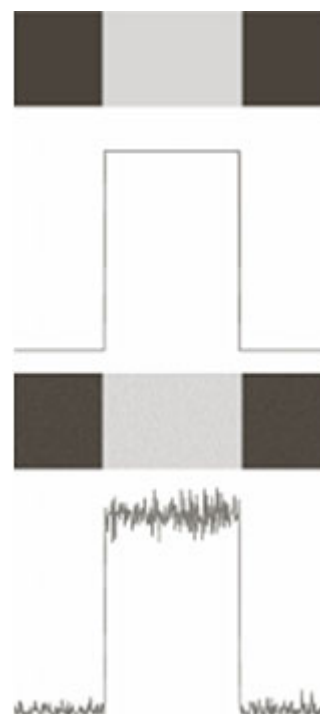
标准差0.0



标准差0.1



一阶导数



二阶导数





一、二阶导数的特性

- 一阶导数通常可以判别图像中较厚的边缘
- 二阶导数对细微细节，如细线、孤立点、噪声等有更强的响应
- 二阶导数在强度上产生斜坡和阶跃过渡的双边缘响应
- 二阶导数的符号可以用来判断一个边缘的转变是由亮变暗还是由暗变亮



锐化空域滤波器: 拉普拉斯算子

拉普拉斯算子是函数(图像) $f(x, y)$ 的二阶各向同性导数算子

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$

旋转不变

$$\frac{\partial^2 f}{\partial x^2} = f(x+1, y) + f(x-1, y) - 2f(x, y)$$

$$\frac{\partial^2 f}{\partial y^2} = f(x, y+1) + f(x, y-1) - 2f(x, y)$$

$$\begin{aligned} \nabla^2 f &= f(x+1, y) + f(x-1, y) + f(x, y+1) + f(x, y-1) \\ &\quad - 4f(x, y) \end{aligned}$$

方向不变性



锐化空域滤波器: 拉普拉斯算子

0	1	0	1	1	1
1	-4	1	1	-8	1
0	1	0	1	1	1

0	-1	0	-1	-1	-1
-1	4	-1	-1	8	-1
0	-1	0	-1	-1	-1

a b c d

FIGURE 3.45 (a) Laplacian kernel used to implement Eq. (3-55). (b) Kernel used to implement an extension of this equation that includes the diagonal terms. (c) and (d) Two other Laplacian kernels.

为什么高通滤波器系数之和等于0

0	-1	0	-1	-1	-1
-1	4	-1	-1	8	-1
0	-1	0	-1	-1	-1

正常使用主观题需2.0以上版本雨课堂

作答



锐化空域滤波器: 拉普拉斯算子

采用拉普拉斯算子进行图像锐化的方式:

$$g(x, y) = f(x, y) + c \left[\nabla^2 f(x, y) \right]$$

where,

$f(x, y)$ is input image, 输入图像

$g(x, y)$ is sharpened images, 输出图像

$c = -1$ if $\nabla^2 f(x, y)$ corresponding to Fig. 3.37(a) or (b)

and $c = 1$ if either of the other two filters is used.

突出强度的不连续性;
强度变化缓慢的区域是不重要的区域;



a b
c d

FIGURE 3.46

(a) Blurred image of the North Pole of the moon.

(b) Laplacian image obtained using the kernel in Fig. 3.45(a).

(c) Image sharpened using Eq. (3-54) with $c = -1$.

(d) Image sharpened using the same procedure, but with the kernel in Fig. 3.45(b).

(Original image courtesy of NASA.)

拉普拉斯图像

锐化图像

锐化图像，不同的kernel

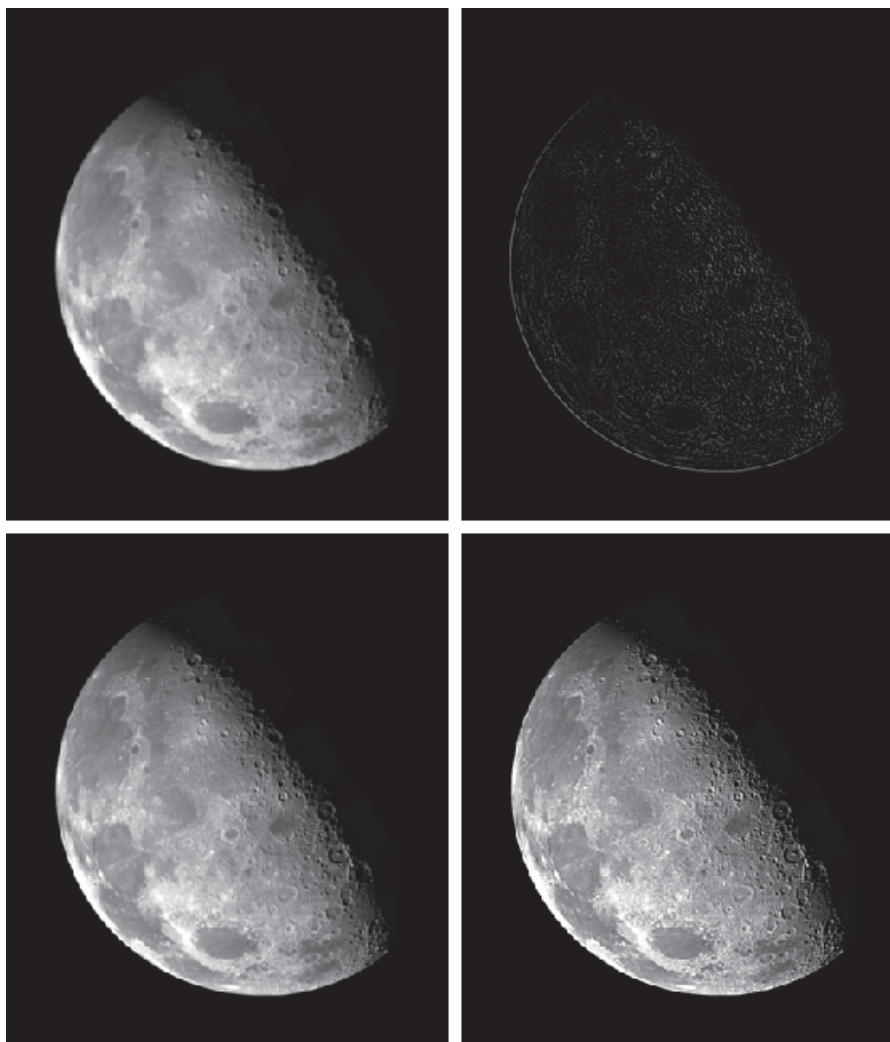


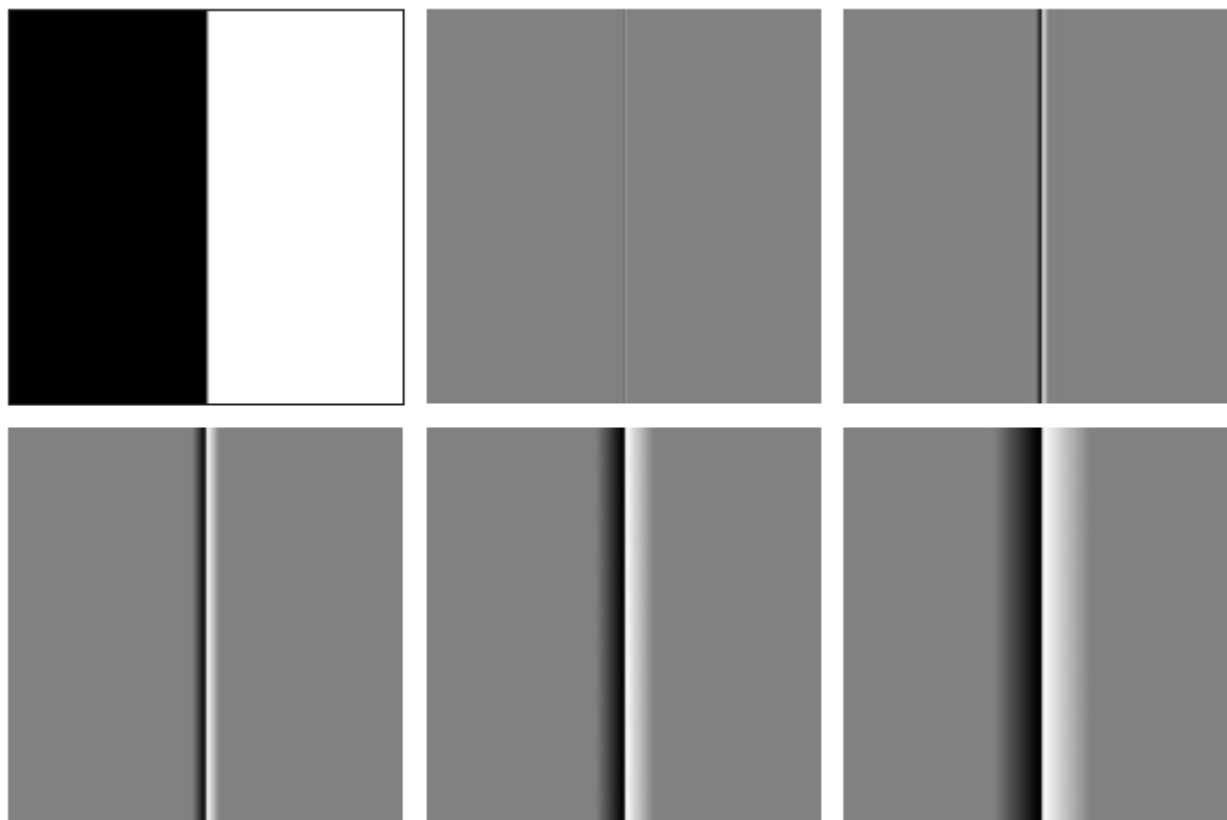
FIGURE 3.47

The Laplacian image from Fig. 3.46(b), scaled to the full $[0, 255]$ range of intensity values. Black pixels correspond to the most negative value in the unscaled Laplacian image, grays are intermediate values, and white pixels correspond to the highest positive value.

图像像素值
scale到
[0,255]



拉普拉斯算子



3×3 , 15×15 , 35×35 , 75×75 , 125×125



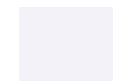
非锐化掩蔽和高提升滤波

- 非锐化掩蔽

锐化图像包括从原始图像中减去图像的非锐化(平滑)形式
例如，印刷出版业

- 步骤

1. 模糊原始图像
2. 从原始图像中减去模糊图像
3. 添加mask到原来的





非锐化掩蔽和高提升滤波

设 $f(x, y)$ 为模糊图像，非锐化掩蔽为：

$$g_{mask}(x, y) = f(x, y) - \bar{f}(x, y)$$

然后将蒙版的加权部分加回到原来的：

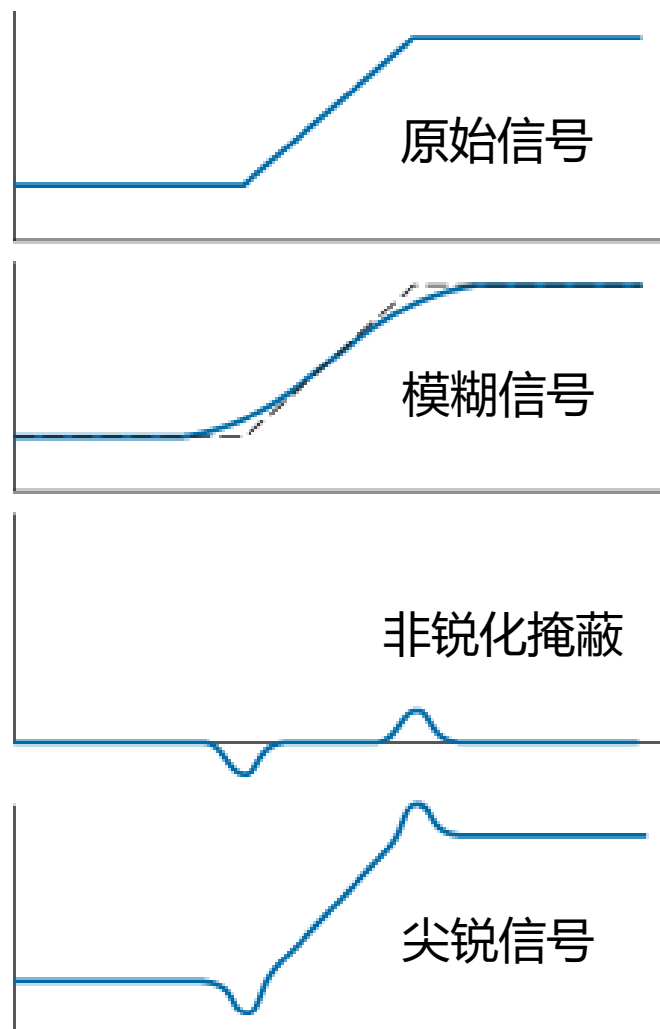
$$g(x, y) = f(x, y) + k * g_{mask}(x, y) \quad k \geq 0$$

当 $k > 1$ 时，该过程称为高增益滤波。



非锐化掩蔽:演示

二阶导数



a
b
c
d

图3.39
非锐化掩蔽力学的一维图解。
(a)原始信号。
(b)信号模糊，原始信号以虚线显示，以供参考。(c)不锋利的口罩。
(d)锐化信号，由c与a相加得到。



非锐化掩蔽和高提升滤波：示例

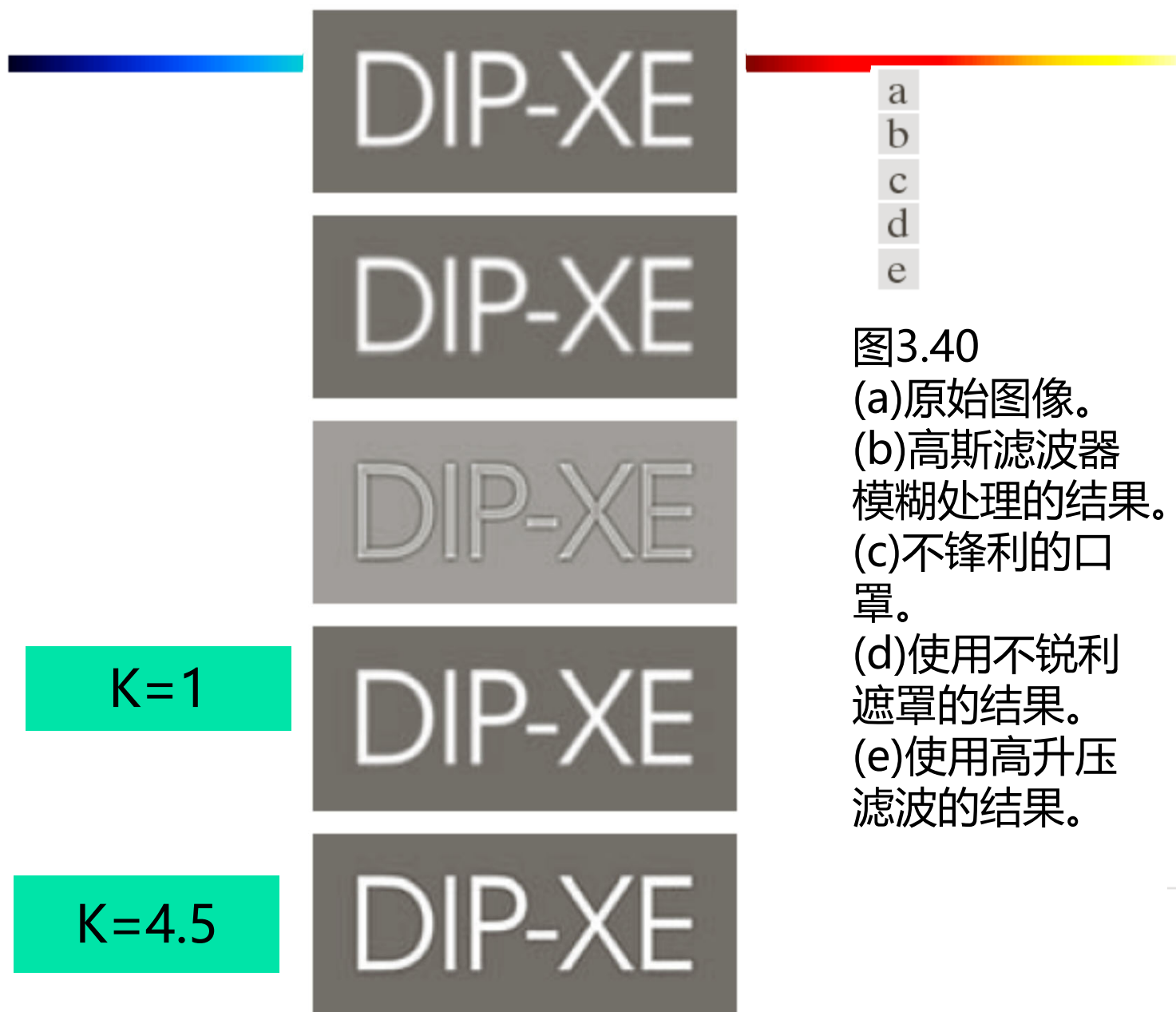
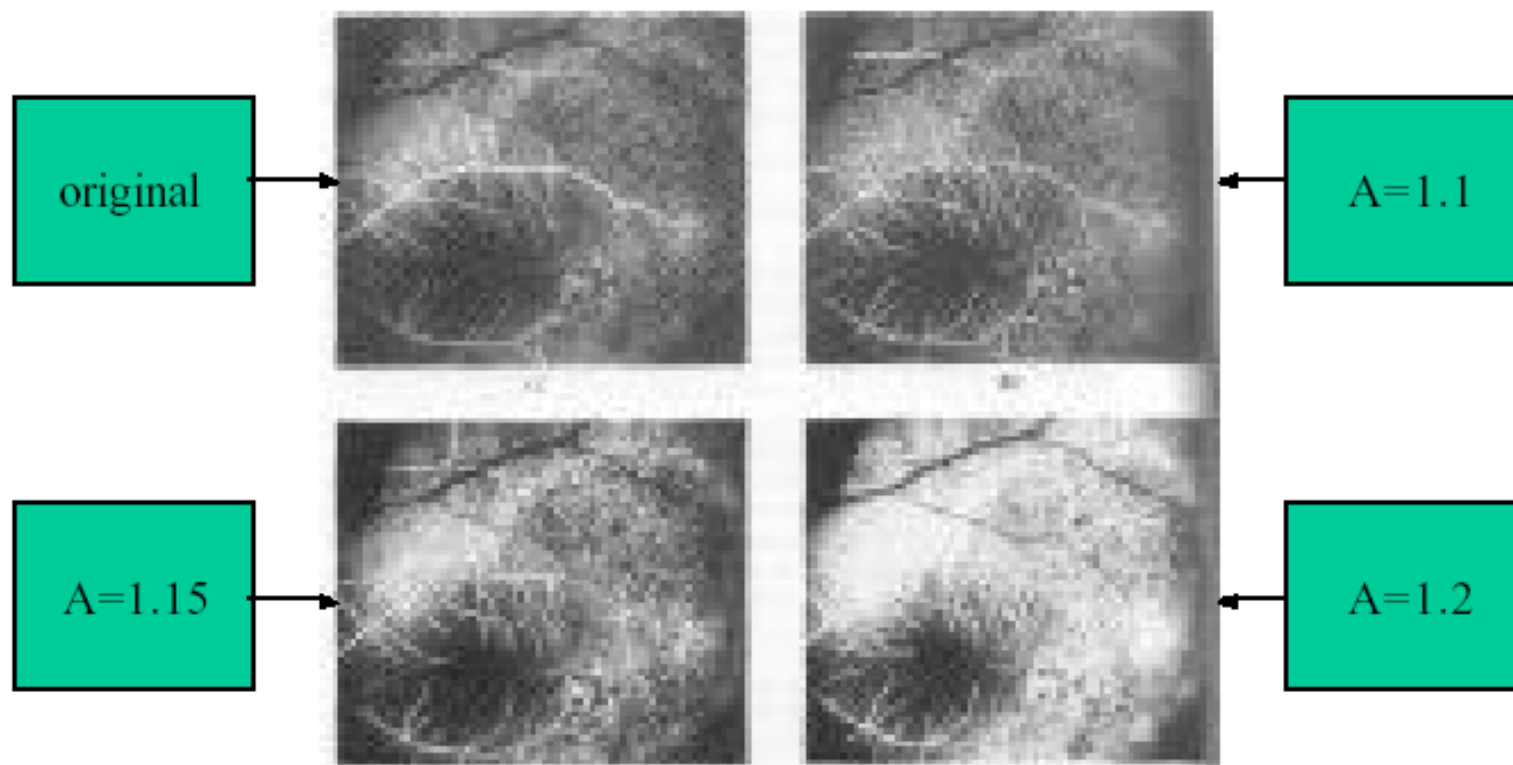


图3.40

- (a)原始图像。
- (b)高斯滤波器模糊处理的结果。
- (c)不锋利的口罩。
- (d)使用不锐利遮罩的结果。
- (e)使用高升压滤波的结果。



高提升滤波：插图

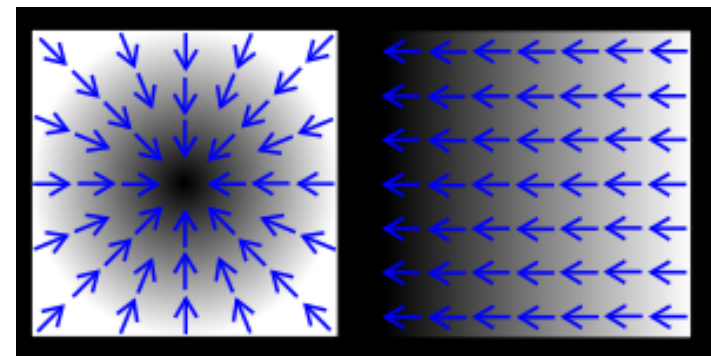




基于一阶导数的图像锐化

对于函数 $f(x, y)$ ，在坐标 (x, y) 处的梯度定义为：

$$\nabla f \equiv \text{grad}(f) \equiv \begin{bmatrix} g_x \\ g_y \end{bmatrix} = \begin{bmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{bmatrix}$$



向量 ∇f 的大小，记为 $M(x, y)$

梯度图像

$$M(x, y) = \text{mag}(\nabla f) = \sqrt{g_x^2 + g_y^2}$$

∇f 的方向

$$\alpha(x, y) = \tan^{-1} \left[\frac{g_x}{g_y} \right]$$



基于一阶导数的图像锐化

向量 ∇f 的大小, 记为 $M(x, y)$

$$M(x, y) = \text{mag}(\nabla f) = \sqrt{g_x^2 + g_y^2}$$

各向同性

$$M(x, y) \approx |g_x| + |g_y|$$

非线性

z_1	z_2	z_3
z_4	z_5	z_6
z_7	z_8	z_9

$$M(x, y) = |z_8 - z_5| + |z_6 - z_5|$$



基于一阶导数的图像锐化

Roberts交叉梯度算子

$$M(x, y) \approx |z_9 - z_5| + |z_8 - z_6|$$

Sobel 算子

z_1	z_2	z_3
z_4	z_5	z_6
z_7	z_8	z_9

$$M(x, y) \approx |(z_7 + 2z_8 + z_9) - (z_1 + 2z_2 + z_3)| \\ + |(z_3 + 2z_6 + z_9) - (z_1 + 2z_4 + z_7)|$$



更多梯度运算符

- 即使是尺寸也难以使用。近似 ∇f 在 Z_5 处，使用 3×3 邻域:

$$\nabla f \approx |(z_7 + z_8 + z_9) - (z_1 + z_2 + z_3)| + |(z_3 + z_6 + z_9) - (z_1 + z_4 + z_7)|$$

- 第一行和第三行之间的差 $\sim \nabla f$ 在x方向
- 第一列和第三列的差值 $\sim \nabla f$ 在y方向上
- 例如:Prewitt 算子

$$\frac{\partial f}{\partial x}$$

-1	-1	-1
0	0	0
1	1	1

-1	0	1
-1	0	1
-1	0	1

$$\frac{\partial f}{\partial y}$$

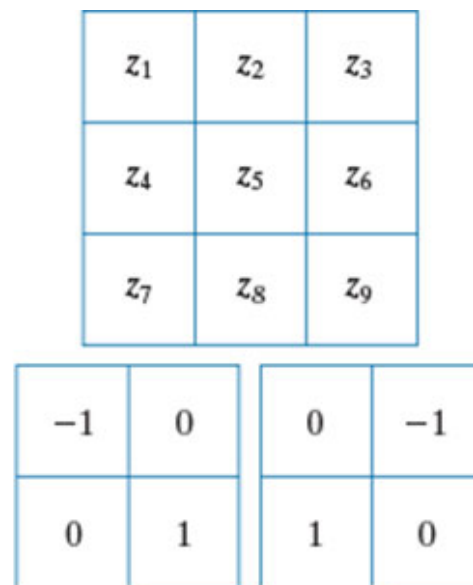
Prewitt



基于一阶导数的 图像锐化

掩码之和为0,
恒定强度无变化

$$\frac{\partial f}{\partial x}$$



Roberts

-1	-1	-1	-1	0	1
0	0	0	-1	0	1
1	1	1	-1	0	1

Prewitt

-1	-2	-1	-1	0	1
0	0	0	-2	0	2
1	2	1	-1	0	1

Sobel

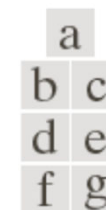


图10.14
图像的3 × 3
区域(z是强度
值)和用于计
算标记为z₅的
点的梯度的各
种掩模。

$$\frac{\partial f}{\partial y}$$



示例

消除缓慢变化的背景特征

-1	-2	-1	-1	0	1
0	0	0	-2	0	2
1	2	1	-1	0	1

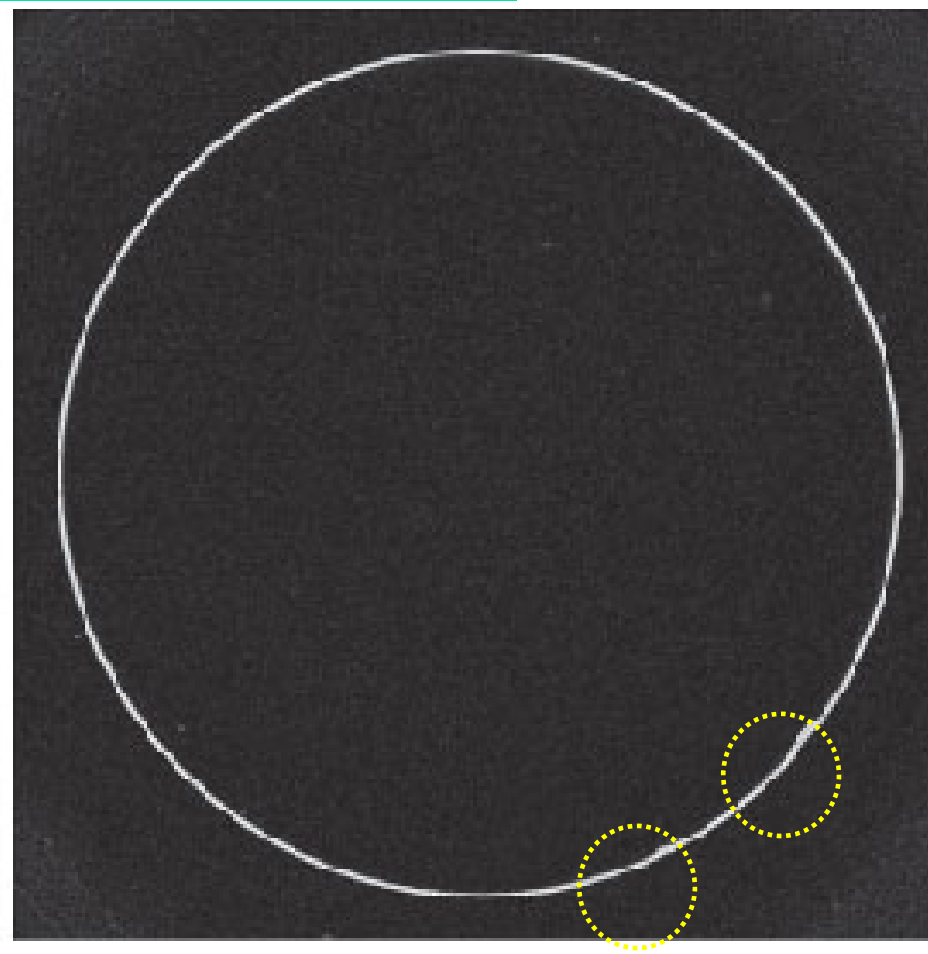
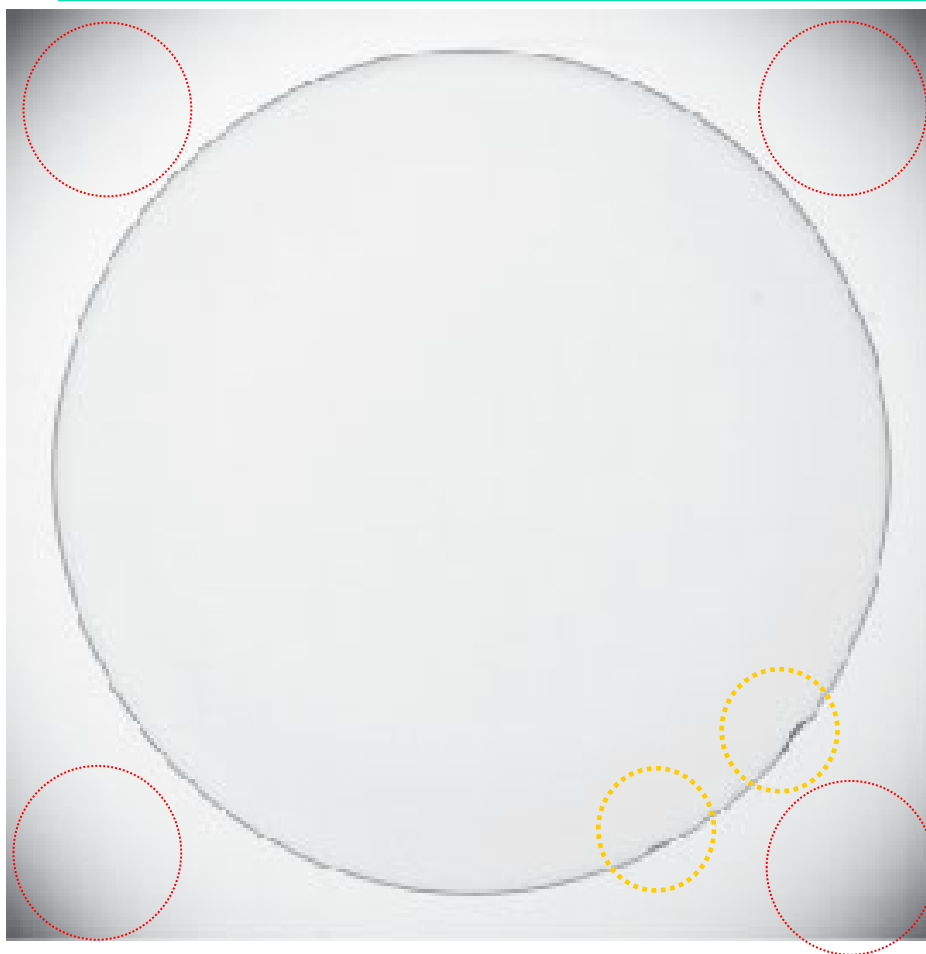
Sobel

a b

FIGURE 3.51

(a) Image of a contact lens (note defects on the boundary at 4 and 5 o'clock).

(b) Sobel gradient. (Original image courtesy of Perceptics Corporation.)





图像与视频处理
Image and Video Processing

边缘检测



a	b
c	d

FIGURE 10.16

(a) Original image of size 834×1114 pixels, with intensity values scaled to the range $[0, 1]$.

(b) $|g_x|$, the component of the gradient in the x -direction, obtained using the Sobel mask in Fig. 10.14(f) to filter the image.

(c) $|g_y|$, obtained using the mask in Fig. 10.14(g).

(d) The gradient image, $|g_x| + |g_y|$.



边缘检测

-1	-2	-1	-1	0	1
0	0	0	-2	0	2
1	2	1	-1	0	1

Sobel

a b
c d

FIGURE 10.16

(a) Original image of size 834×1114 pixels, with intensity values scaled to the range $[0, 1]$.

(b) $|g_x|$, the component of the gradient in the x -direction, obtained using the Sobel mask in Fig. 10.14(f) to filter the image.

(c) $|g_y|$, obtained using the mask in Fig. 10.14(g).

(d) The gradient image, $|g_x| + |g_y|$.

$|g_x|$





图像与视频处理
Image and Video Processing

边缘检测

-1	-2	-1	-1	0	1
0	0	0	-2	0	2
1	2	1	-1	0	1

Sobel

a	b
c	d

$$|g_y|$$



FIGURE 10.16

(a) Original image of size 834×1114 pixels, with intensity values scaled to the range $[0, 1]$.
(b) $|g_x|$, the component of the gradient in the x -direction, obtained using the Sobel mask in Fig. 10.14(f) to filter the image.
(c) $|g_y|$, obtained using the mask in Fig. 10.14(g).
(d) The gradient image, $|g_x| + |g_y|$.



图像与视频处理
Image and Video Processing

边缘检测-sobel

-1	-2	-1	-1	0	1
0	0	0	-2	0	2
1	2	1	-1	0	1

Sobel

FIGURE 10.16

(a) Original image of size 834×1114 pixels, with intensity values scaled to the range $[0, 1]$.
 (b) $|g_x|$, the component of the gradient in the x -direction, obtained using the Sobel mask in Fig. 10.14(f) to filter the image.
 (c) $|g_y|$, obtained using the mask in Fig. 10.14(g).
 (d) The gradient image, $|g_x| + |g_y|$.

$$|g_x| + |g_y|$$





图像与视频处理
Image and Video Processing

梯度角

-1	-2	-1	-1	0	1
0	0	0	-2	0	2
1	2	1	-1	0	1

Sobel

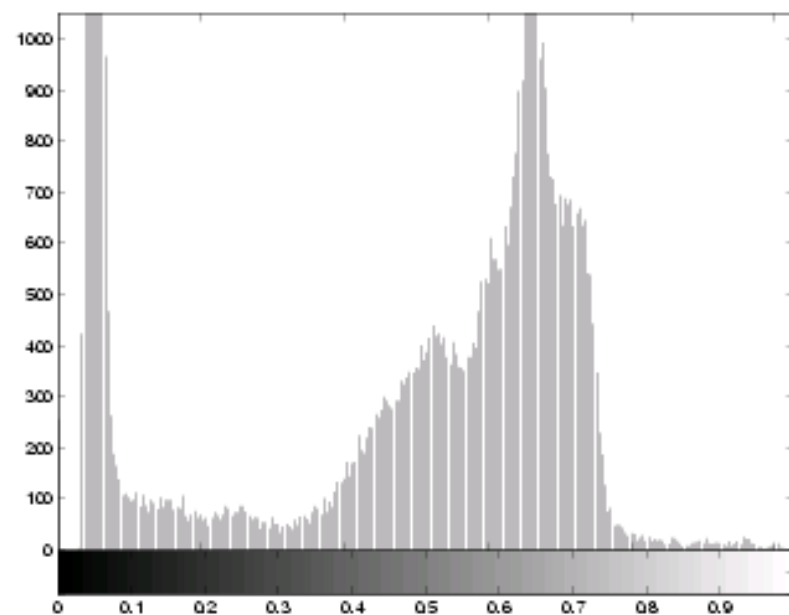


FIGURE 10.17
Gradient angle
image computed
using
Eq. (10.2-11).
Areas of constant
intensity in this
image indicate
that the direction
of the gradient
vector is the same
at all the pixel
locations in those
regions.



图像与视频处理
Image and Video Processing

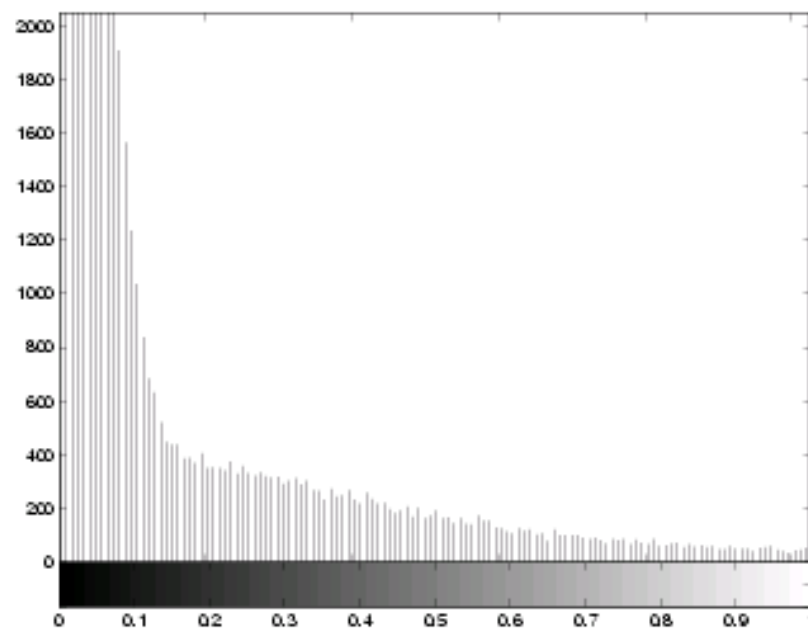
原始图片和直方图





图像与视频处理
Image and Video Processing

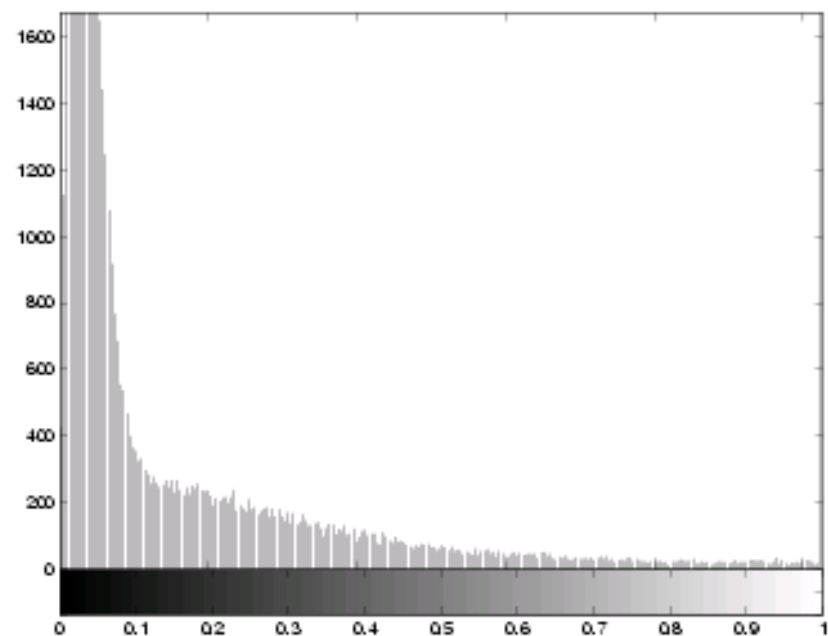
Sobel和直方图





图像与视频处理
Image and Video Processing

Prewitt和直方图





图像与视频处理
Image and Video Processing

低通空间滤波



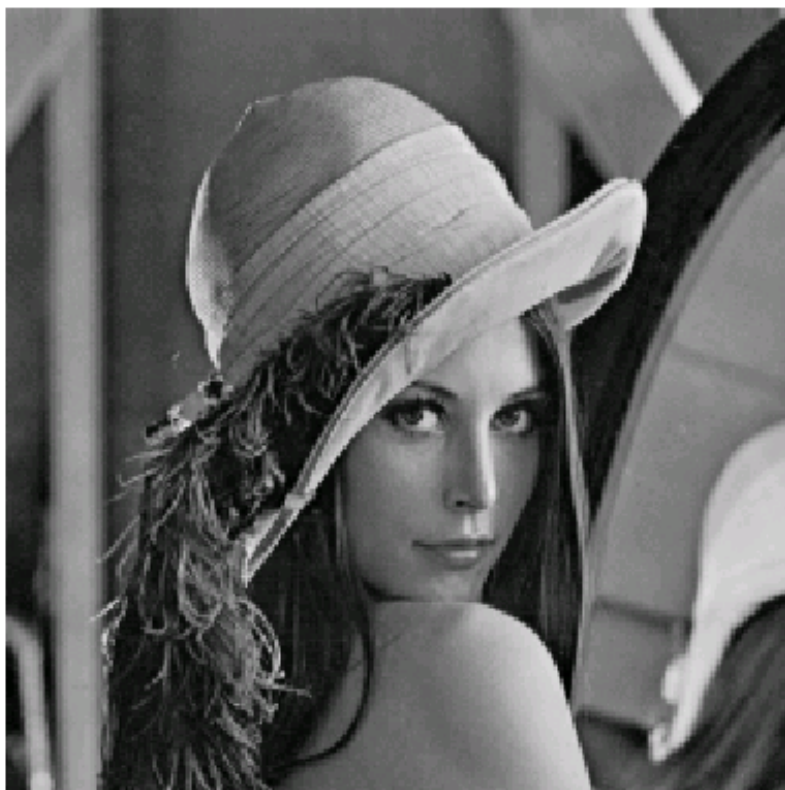
(b) 3×3 mask



(a) 5×5 mask



高通空间滤波原始图像



(a) Original "Lena" image



(a) "Lena" image with 10 dB Gaussian noise



图像与视频处理
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高通空间滤波



(a) Highpass-filtered "Lena"



(b) Highpass-filtered "Lena" with 10 dB noise



图像与视频处理
Image and Video Processing

高升压滤波



(a) $A=1.15$



(b) $A=1.2$



a b
c d

FIGURE 3.52

Transfer functions of ideal 1-D filters in the frequency domain (u denotes frequency).

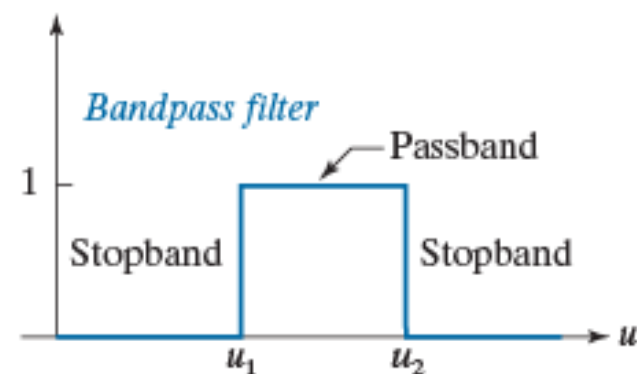
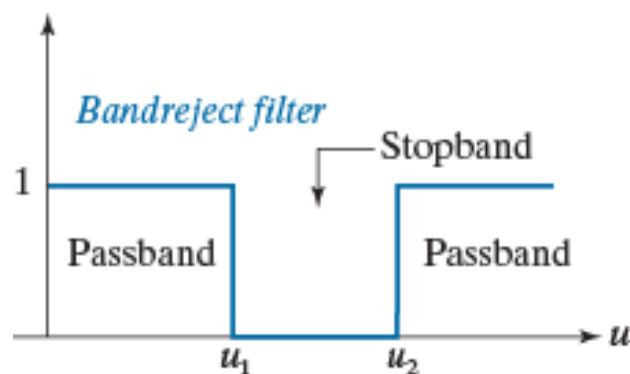
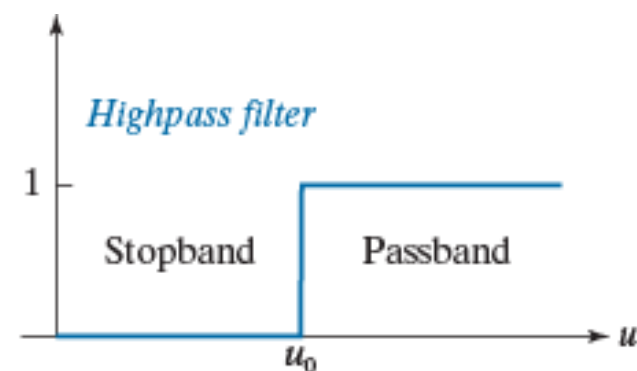
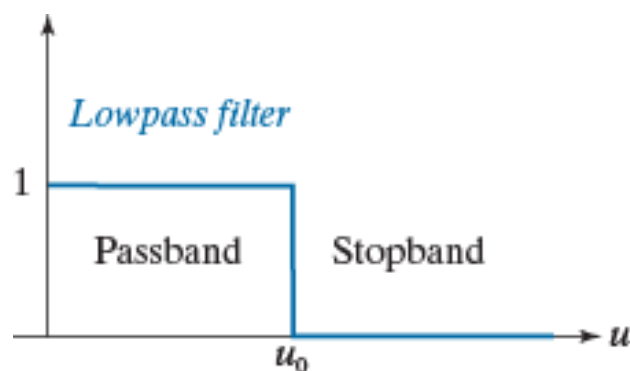
(a) Lowpass filter.

(b) Highpass filter.

(c) Bandreject filter.

(d) Bandpass filter.

(As before, we show only positive frequencies for simplicity.)

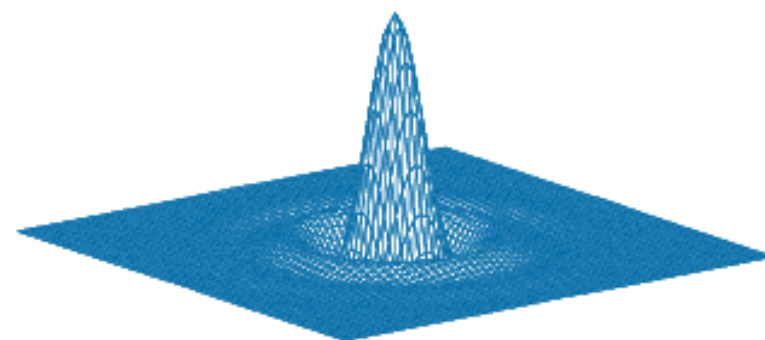
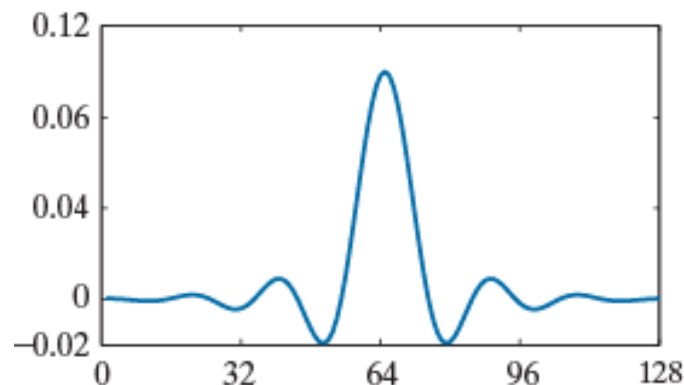




a b

FIGURE 3.54

(a) A 1-D spatial lowpass filter function. (b) 2-D kernel obtained by rotating the 1-D profile about its center.



a b

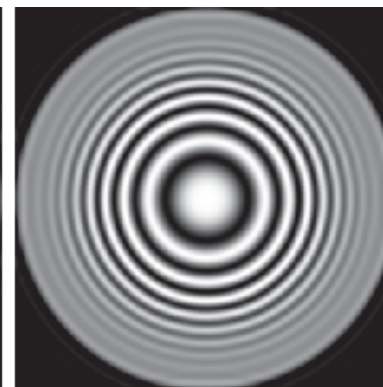
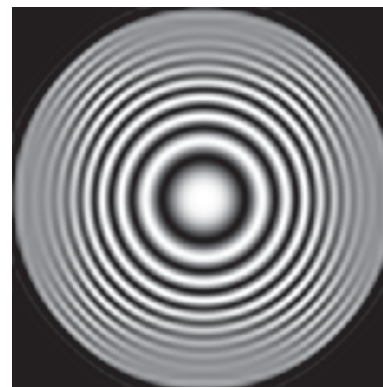
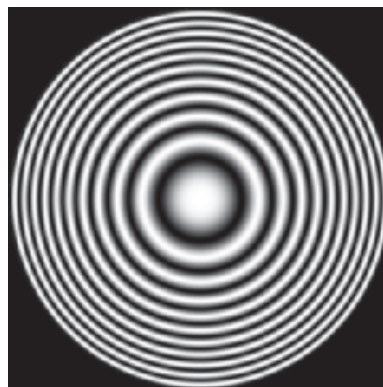
$$z(x, y) = \frac{1}{2} \left[1 + \cos(x^2 + y^2) \right]$$

FIGURE 3.55

(a) Zone plate image filtered with a separable lowpass kernel. (b) Image filtered with the isotropic lowpass kernel in Fig. 3.54(b).

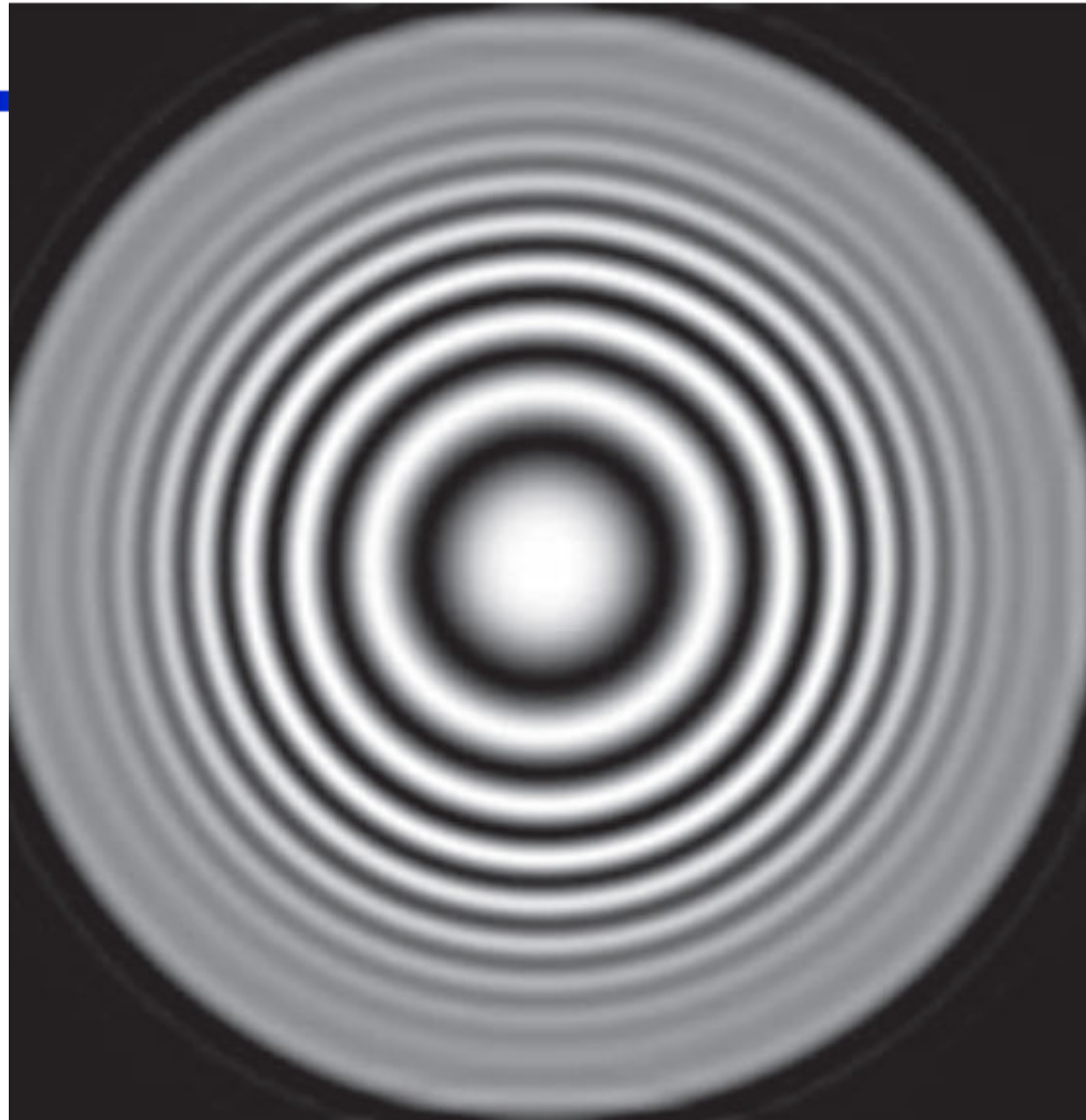
FIGURE 3.53

A zone plate image of size 597 × 597 pixel





Lowpass
result



a	b	c
d	e	f

FIGURE 3.56

Spatial filtering of the zone plate image. (a) Lowpass result; this is the same as Fig. 3.55(b). (b) Highpass result. (c) Image (b) with intensities scaled. (d) Bandreject result. (e) Bandpass result. (f) Image (e) with intensities scaled.



**Highpass
result**

a	b	c
d	e	f

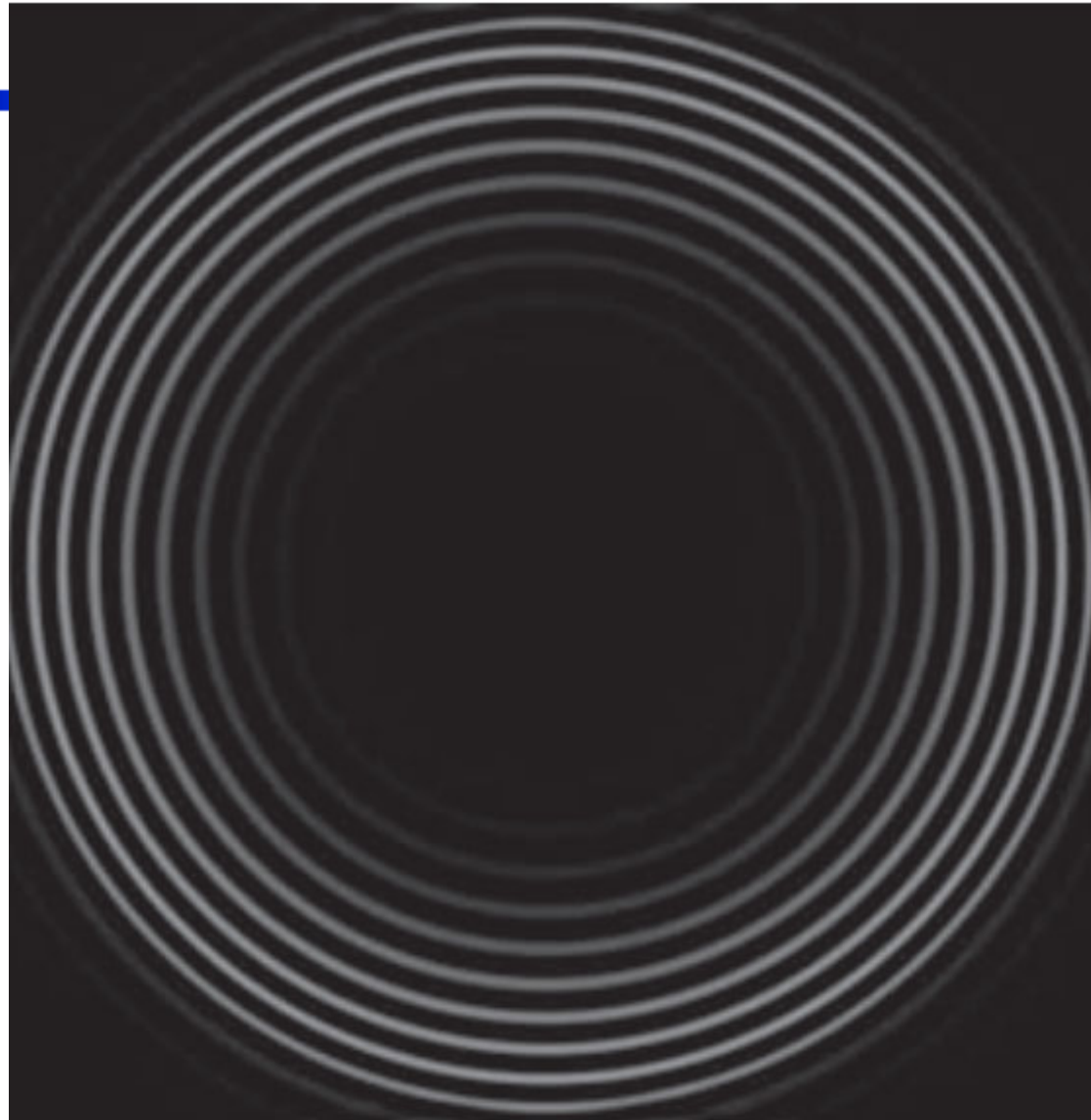


FIGURE 3.56

Spatial filtering of the zone plate image. (a) Lowpass result; this is the same as Fig. 3.55(b). (b) Highpass result. (c) Image (b) with intensities scaled. (d) Bandreject result. (e) Bandpass result. (f) Image (e) with intensities scaled.



**Image (b)
with
intensities
scaled**

a	b	c
d	e	f

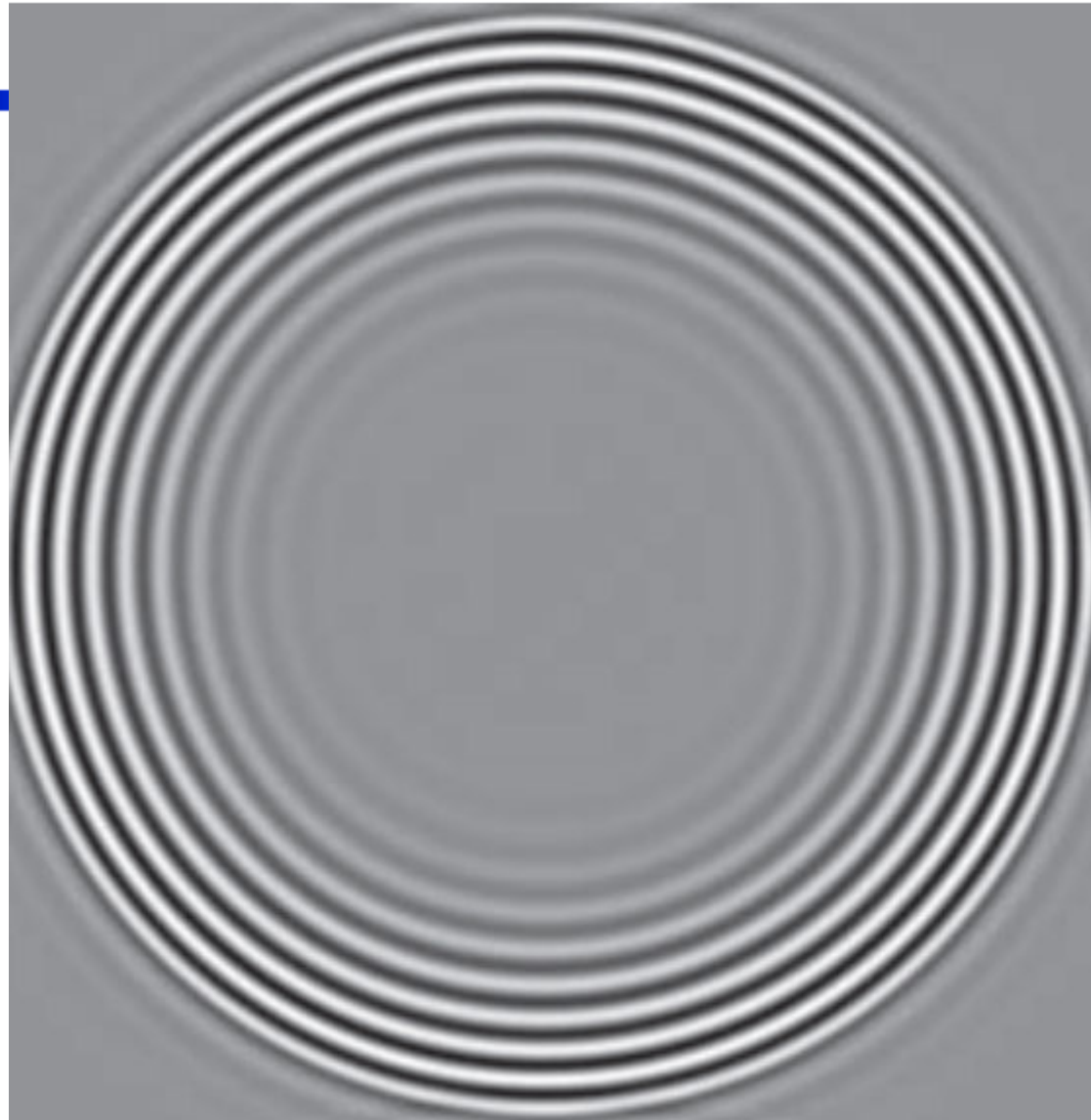
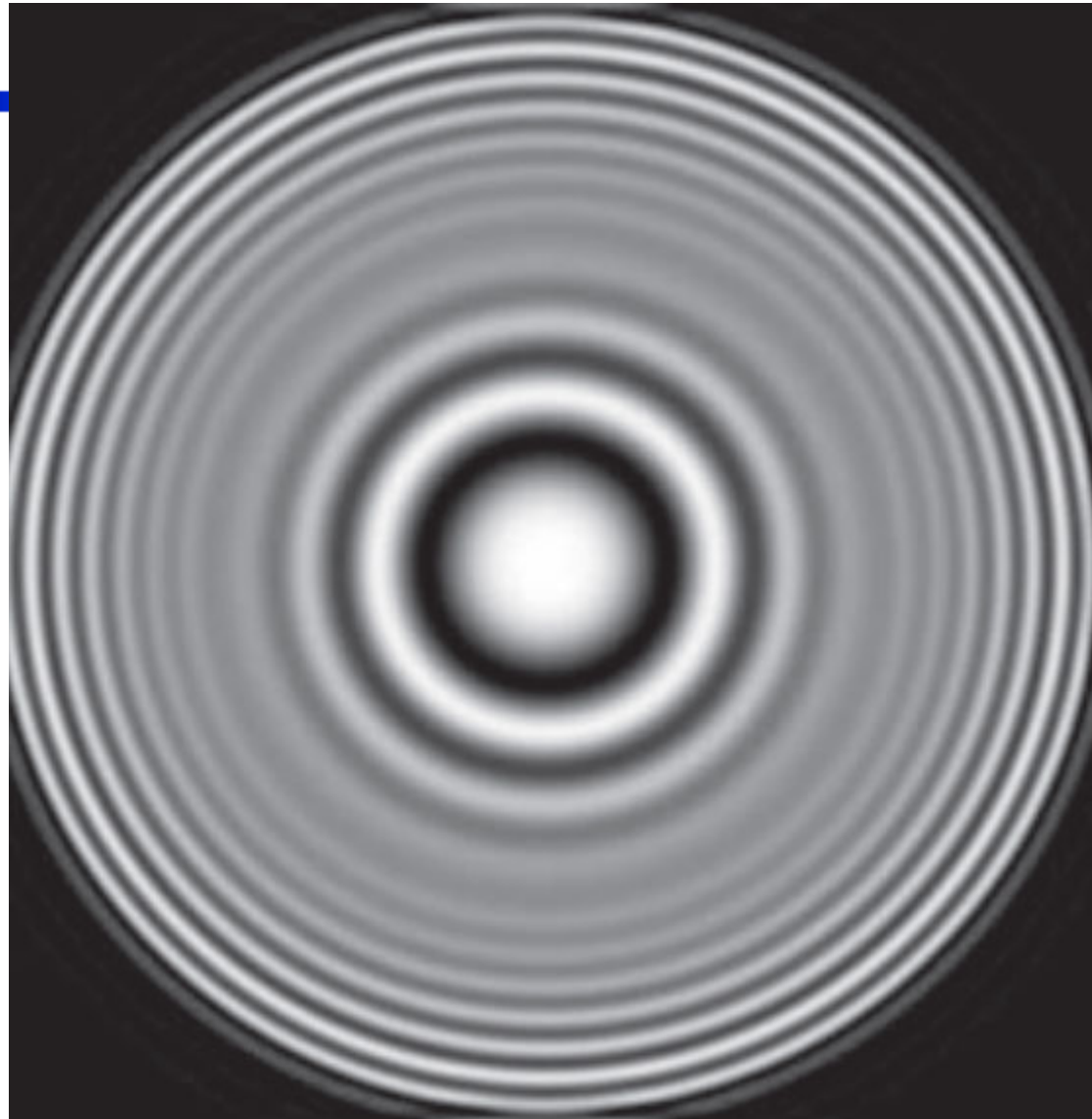


FIGURE 3.56

Spatial filtering of the zone plate image. (a) Lowpass result; this is the same as Fig. 3.55(b). (b) Highpass result. (c) Image (b) with intensities scaled. (d) Bandreject result. (e) Bandpass result. (f) Image (e) with intensities scaled.



**Banreject
result**



a	b	c
d	e	f

FIGURE 3.56

Spatial filtering of the zone plate image. (a) Lowpass result; this is the same as Fig. 3.55(b). (b) Highpass result. (c) Image (b) with intensities scaled. (d) Bandreject result. (e) Bandpass result. (f) Image (e) with intensities scaled.



**Bandpass
result**

a	b	c
d	e	f

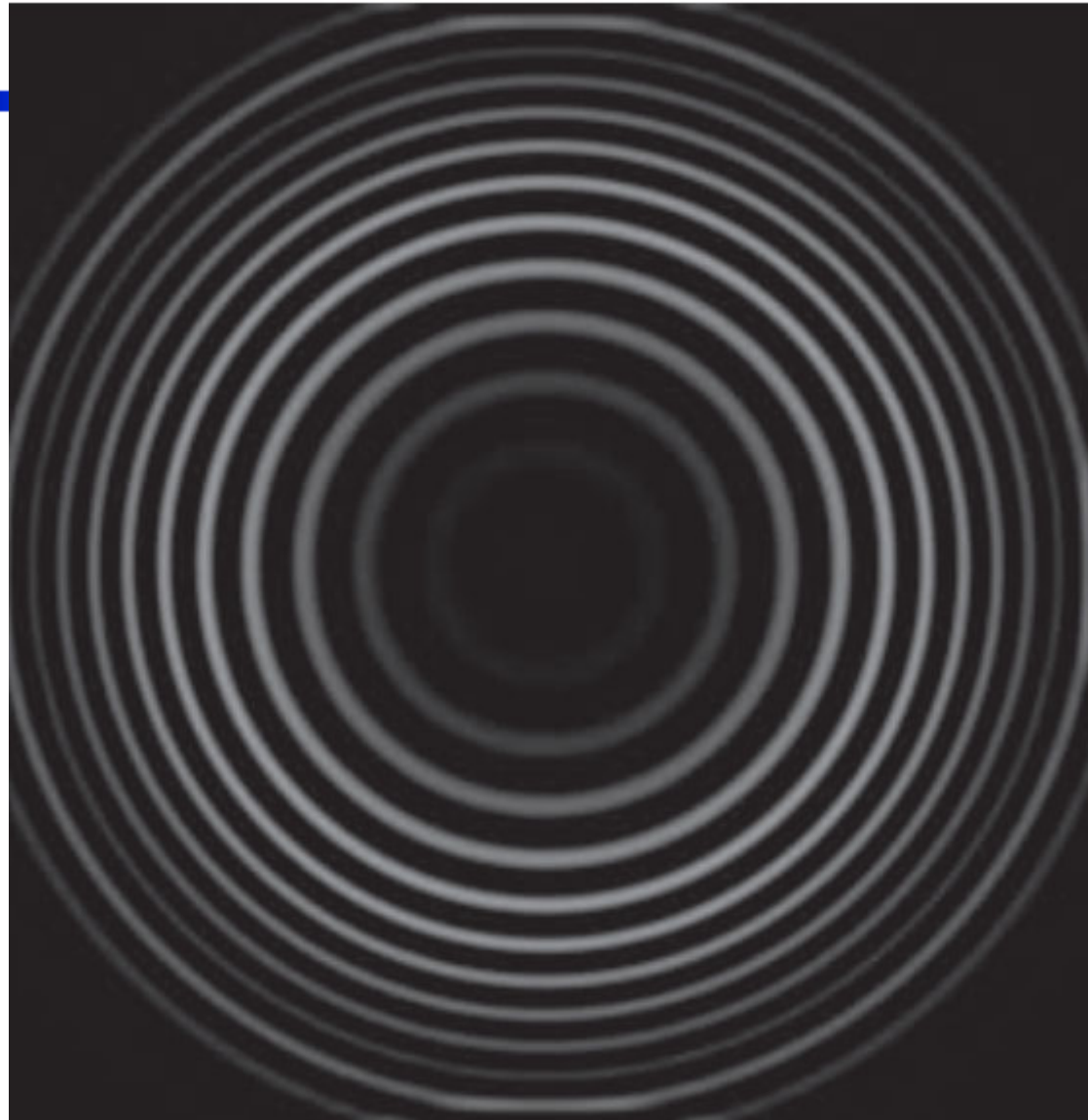


FIGURE 3.56

Spatial filtering of the zone plate image. (a) Lowpass result; this is the same as Fig. 3.55(b). (b) Highpass result. (c) Image (b) with intensities scaled. (d) Bandreject result. (e) Bandpass result. (f) Image (e) with intensities scaled.



**Image (e)
with
intensities
scaled**

a	b	c
d	e	f

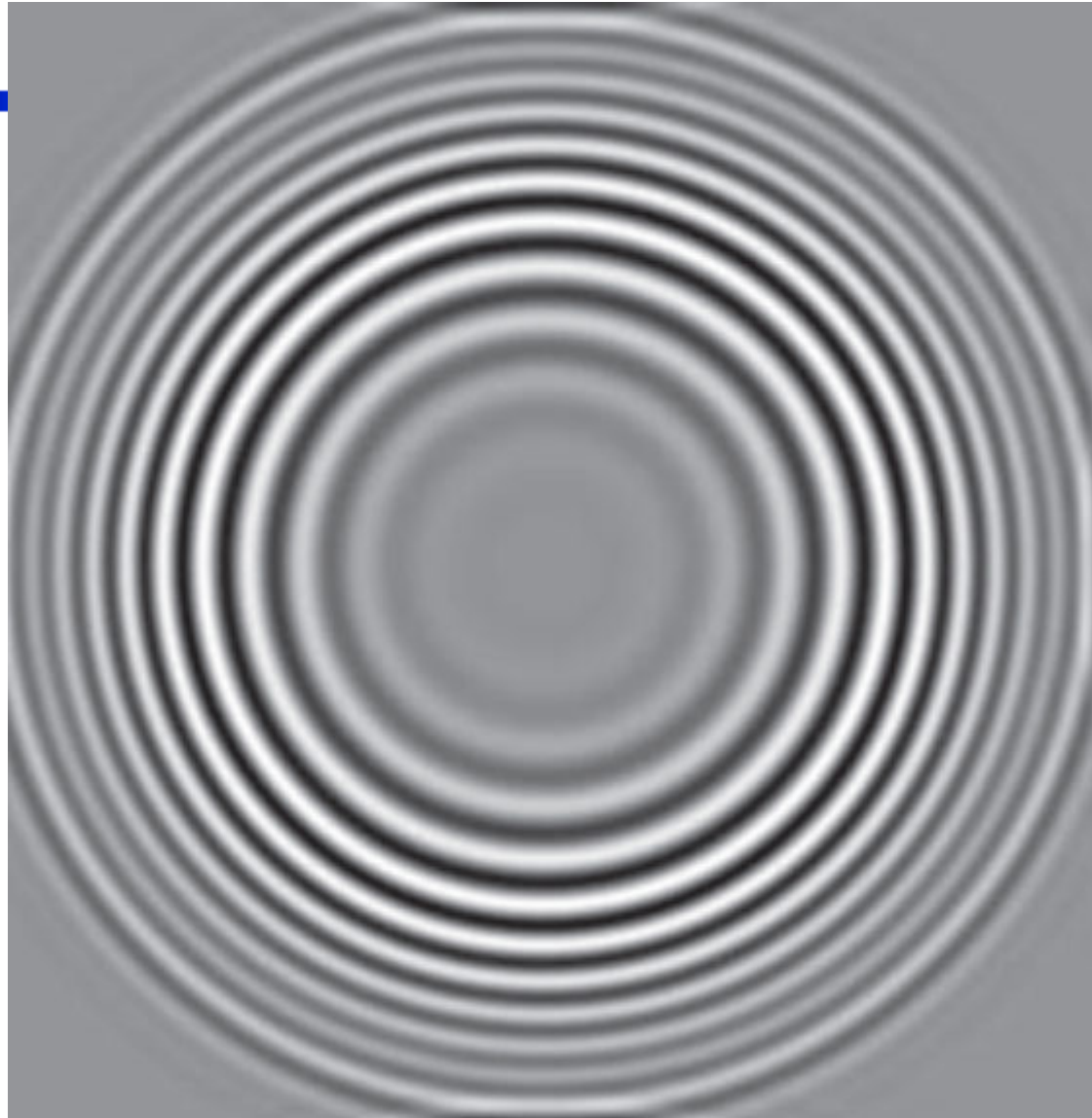


FIGURE 3.56

Spatial filtering of the zone plate image. (a) Lowpass result; this is the same as Fig. 3.55(b). (b) Highpass result. (c) Image (b) with intensities scaled. (d) Bandreject result. (e) Bandpass result. (f) Image (e) with intensities scaled.

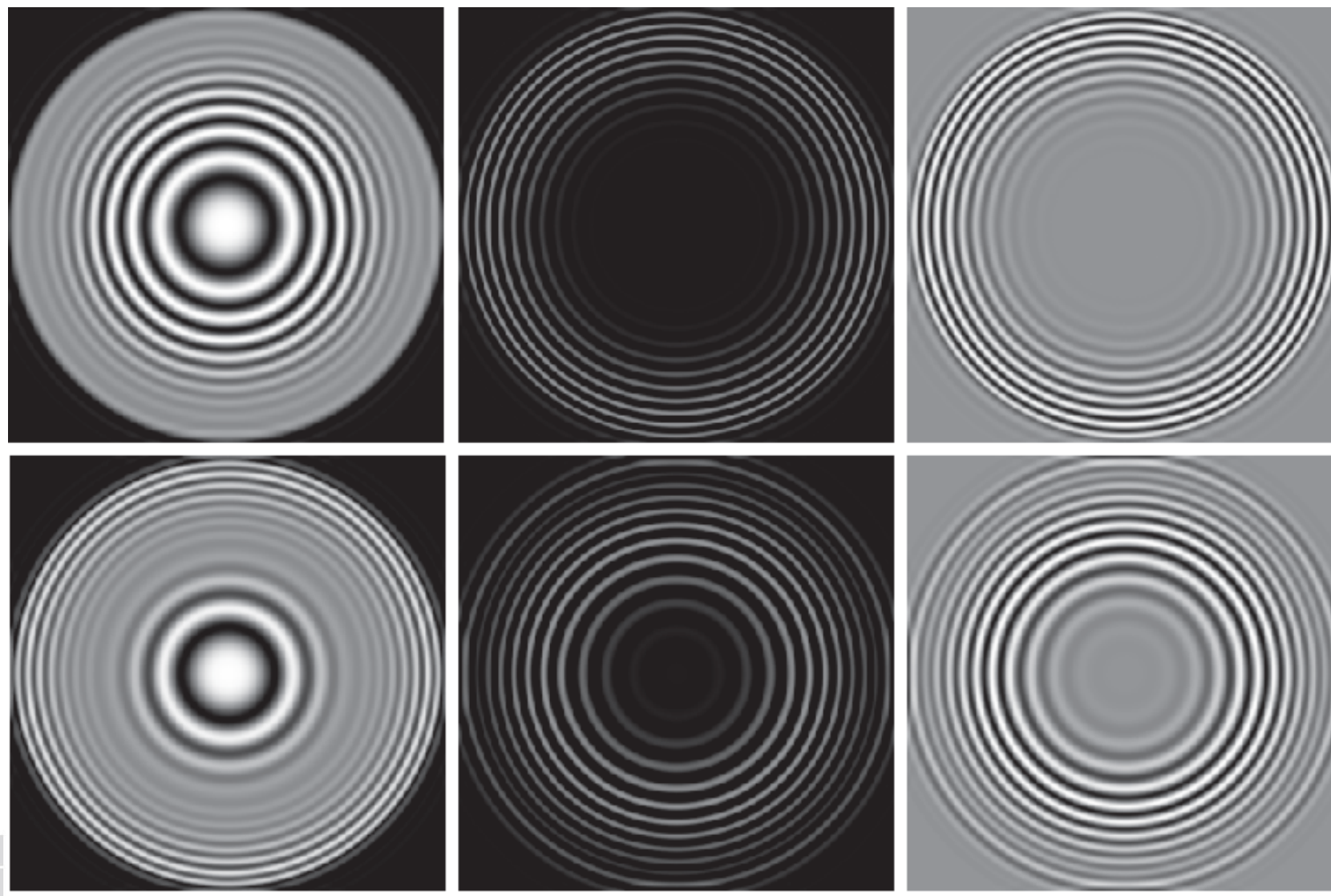


FIGURE 3.56

Spatial filtering of the zone plate image. (a) Lowpass result; this is the same as Fig. 3.55(b). (b) Highpass result. (c) Image (b) with intensities scaled. (d) Bandreject result. (e) Bandpass result. (f) Image (e) with intensities scaled.



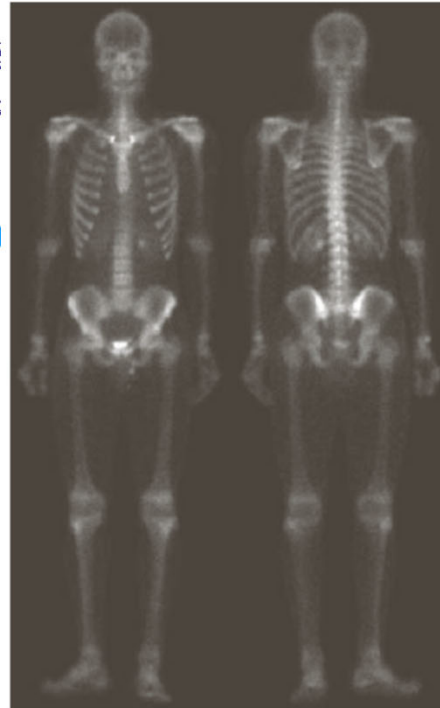
Example: 图像与视频处理 Image and Video Processing

Combining Spatial Enhancement Methods

Goal:

Enhance the
image by
sharpening it
and by
bringing out
more of the
skeletal detail

$$c=a+b$$



Laplacian

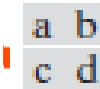


FIGURE 3.57

(a) Image of whole body bone scan.

(b) Laplacian of (a).

(c) Sharpened image obtained by adding (a) and (b).

(d) Sobel gradient of image (a). (Original image courtesy of G.E. Medical Systems.)

Sobel gradient

二阶倒数：
细节，平滑处
一阶导数：
边缘突出



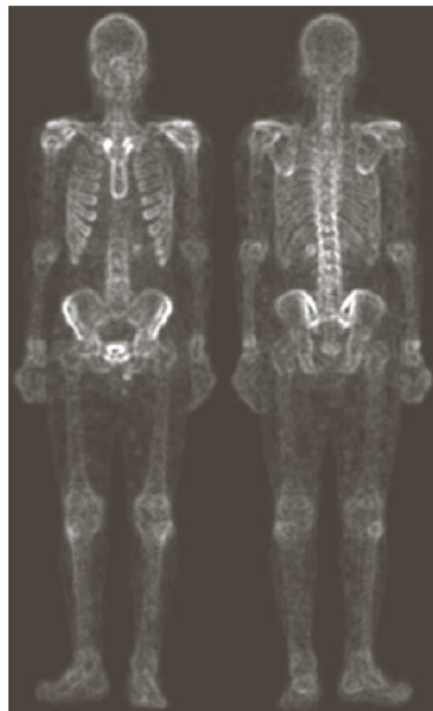
Example:

Combining
Spatial
Enhancement
Methods

Goal: $e=d$
smoothing

Enhance the
image by
sharpening it
and by bringing
out more of the
skeletal detail

$$g=a+f$$



$$F=c*e$$

e f
g h

FIGURE 3.57

(Continued)

(e) Sobel image
smoothed with a
 5×5 box filter.

(f) Mask image
formed by the
product of (b)
and (e).

(g) Sharpened
image obtained
by the adding
images (a) and (f).

(h) Final result
obtained by
applying a power-
law transformation
to (g). Compare
images (g) and (h)
with (a). (Original
image courtesy
of G.E. Medical
Systems.)

$$H=\text{power}(g)$$



平均/中位数图像

- 我们能用这个做什么？





图像与视频处理
Image and Video Processing

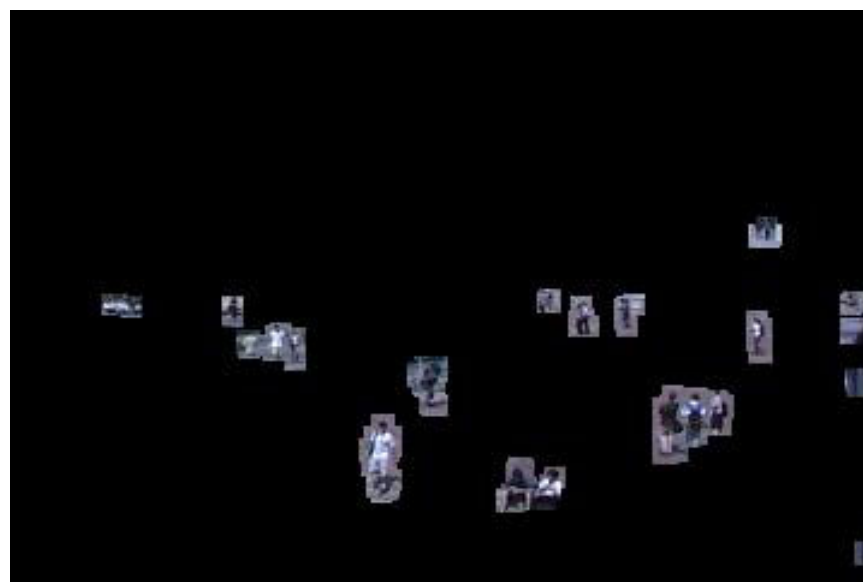
背景消减



-



=





图像与视频处理
Image and Video Processing

群众合成 (与Pooja Nath合作)



1. 在每一帧中做背景减法
2. 寻找并记录 "斑点".
3. 在合成时, 随机取样, 注意不要重叠。



图像与视频处理
Image and Video Processing

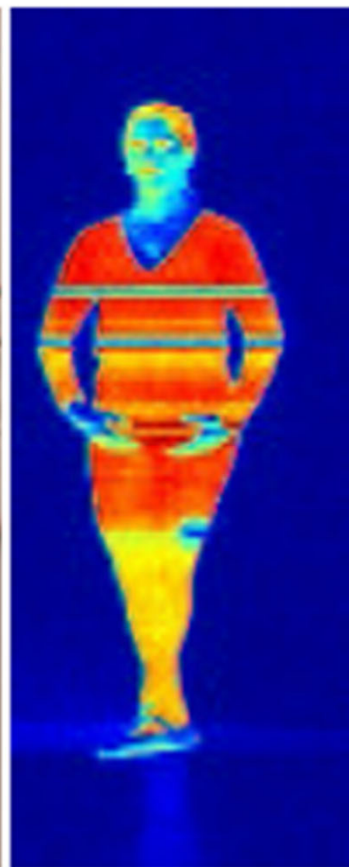
背景消减



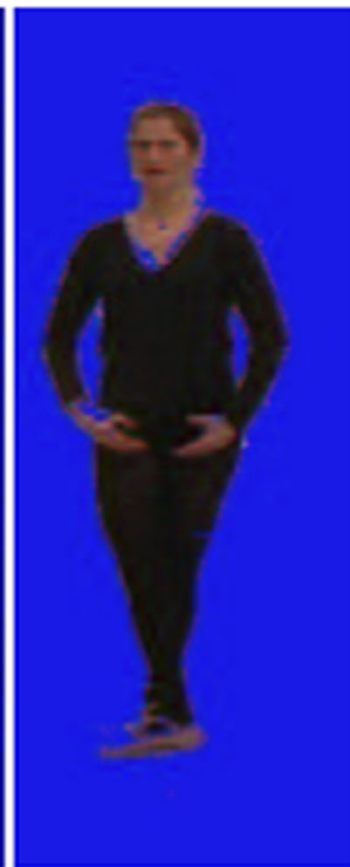
一个视频框架



估计背景



图像区别



蓝色表示
阈值化前景



图像与视频处理
Image and Video Processing

蓝屏





变形=对象平均化



- 其目的是在两个对象之间找到 "平均"。
 - 不是两个物体的平均图像...
 - ...而是一个平均物体的图像!
- 我们怎样才能实现平稳过渡?
 - 在时间上做一个 "加权平均" t
 - 我们怎么知道平均物体是什么样子的?
 - 我们没有任何头绪!
 - 但我们通常可以伪造一些合理的东西
- 通常需要用户/艺术家的输入





图像与视频处理
Image and Video Processing

对齐，然后交叉拆分



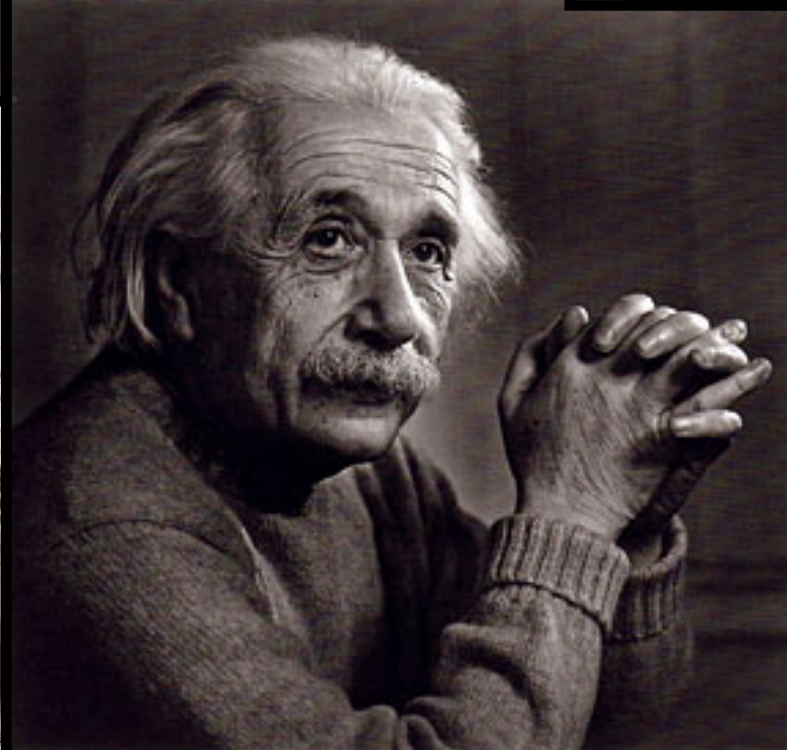
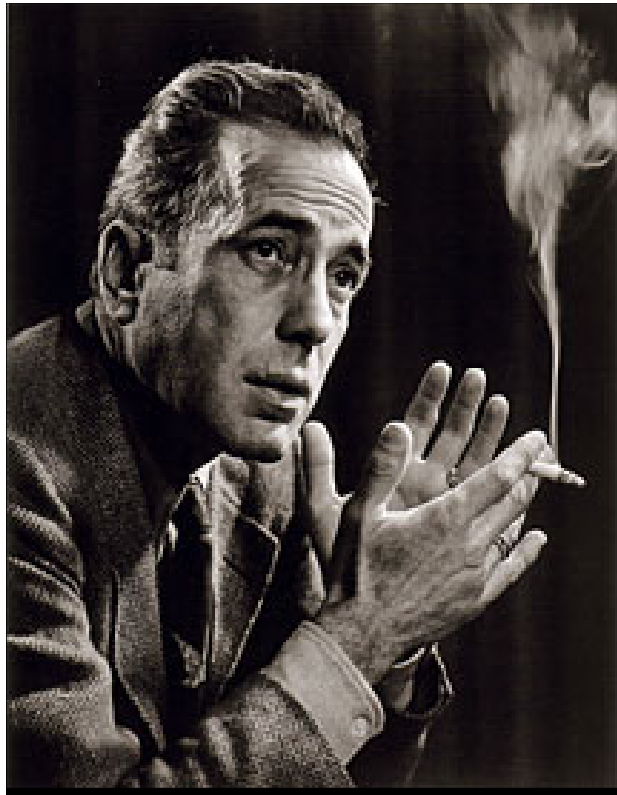
- 先对齐，再交叉溶解
 - 使用全局卷曲进行对齐 - 图片仍然有效



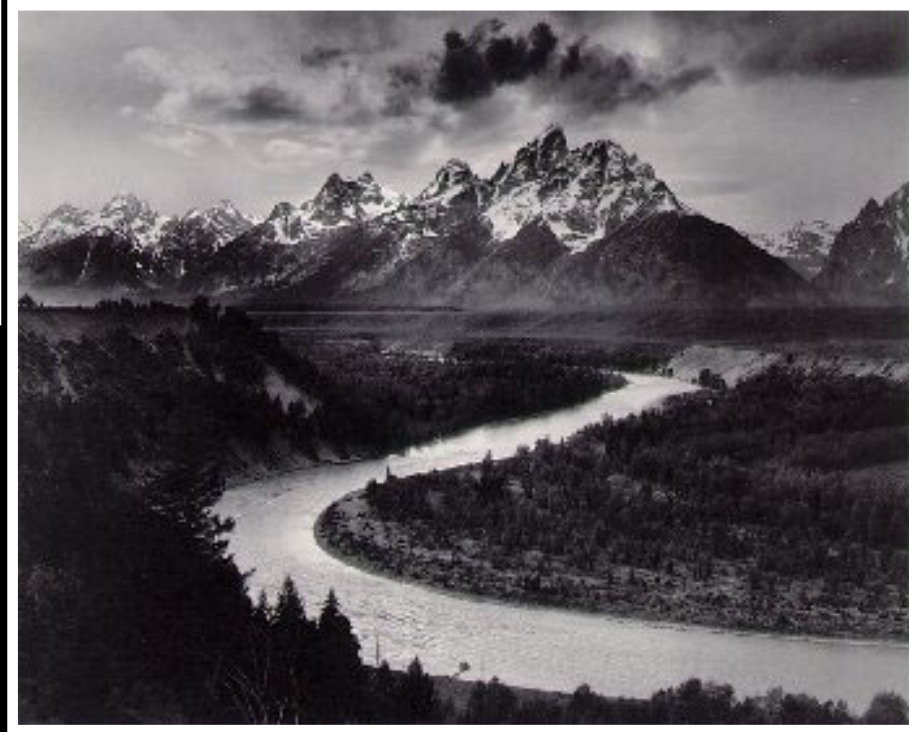
图像与视频处理
Image and Video Processing

画家 (图片)





Ansel Adams



摄影不仅仅是一种事实性的思想交流的媒介，
它是一种创造性的艺术。







图像与视频处理
Image and Video Processing

以上是全部内容





目录

- 强度转换
 - 图像负片
 - 对数转换
 - 伽马变换
 - 碎片化线性变换
- 直方图处理
 - 直方图均衡化
 - 直方图匹配
 - 局部直方图处理
- 空间滤波器
 - 空间相关性和卷积
 - 平滑线性滤波器和空间掩码
 - 非线性滤波器
 - 锐化滤波器



作业

Project3:直方图图像增强;

共10幅经变亮或者变暗处理的源图像;

要求:

- 1.把附件图像的直方图画出;
- 2.把所有图像进行直方图均衡; 输出均衡后的图像和源图像进行比对;
分析改善内容;
- 3.进一步把图像按照对源图像直方图的观察, 各自指定直方图, 进行直方图匹配, 进行图像增强;

要求: 按照作业报告格式提交报告, 下一双周-周四提交;
另附原始图像4幅:



Project 4

- 空域低通滤波器：分别用高斯滤波器和中值滤波器去平滑测试图像test1和2，模板大小分别是 3×3 ， 5×5 ， 7×7 ；
 - 利用固定方差 $\sigma=1.5$ 产生高斯滤波器. 附件有产生高斯滤波器的方法；
- 利用高通滤波器滤波测试图像test3,4：包括 unsharp masking, Sobel edge detector, and Laplace edge detection.
- 按标准格式提交报告；